Implementation of a Normalized Database in MySQL

The Pressly Ad Agency — Advertising Database

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Objectives

- Create and select a database in MySQL.
- Create tables using suitable data types and integrity constraints.
- Define primary keys, composite keys, and foreign keys.
- Insert sample data in the correct dependency order.
- Retrieve data using SELECT statements.
- Combine related tables using JOIN operations.
- Verify that the normalized tables can reconstruct the original information.

AD CAMPAING MIX with Before Normalization (with Update Anomalies)

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	**AutmnStyle20**	**6.9.2020**	**5 days**	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%
????	????	????	????	????	????	**7**	**Internet**	**National**	**????**

Deletion Anomaly Example:
Cannot delete campaign 444 without also deleting all the data about campaign manager CM103 and campaign modus 6.

Modification Anomaly Example:
To change the duration of the campaign 222 from 30 to 45 days, three records have to be modified.

Insertion Anomaly Example:
Cannot insert new campaign modus 7 without inserting an actual campaign using the new modus 7.

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AD CAMPAING MIX After Normalisation to BCNF

AD CAMPAIGN

AdCampaignID	AdCampaignName	StartDate	Duration	CampaignMgrID
111	SummerFun20	6.6.2020	12 days	CM100
222	SummerZing20	6.8.2020	45 days	CM101
333	FallBall20	6.9.2020	12 days	CM102
555	AutmnColors20	6.9.2020	3 days	CM100

Modification Anomaly Resolved:
Only one record modified.

CAMPAIGN MANAGER

CampaignMgrID	CampaignMgrName
CM100	Roberta
CM101	Sue
CM102	John
CM103	Nancy

MODUS

ModusID	Media	Range
1	TV	Local
2	TV	National
3	Radio	Local
4	Radio	National
5	Print	Local
6	Print	National
7	Internet	National

Deletion Anomaly Resolved:
Campaign 444 deleted but all the data about the campaign manager CM103 and the campaign modus 6 remain in the database.

AD CAMPAIGN MIX

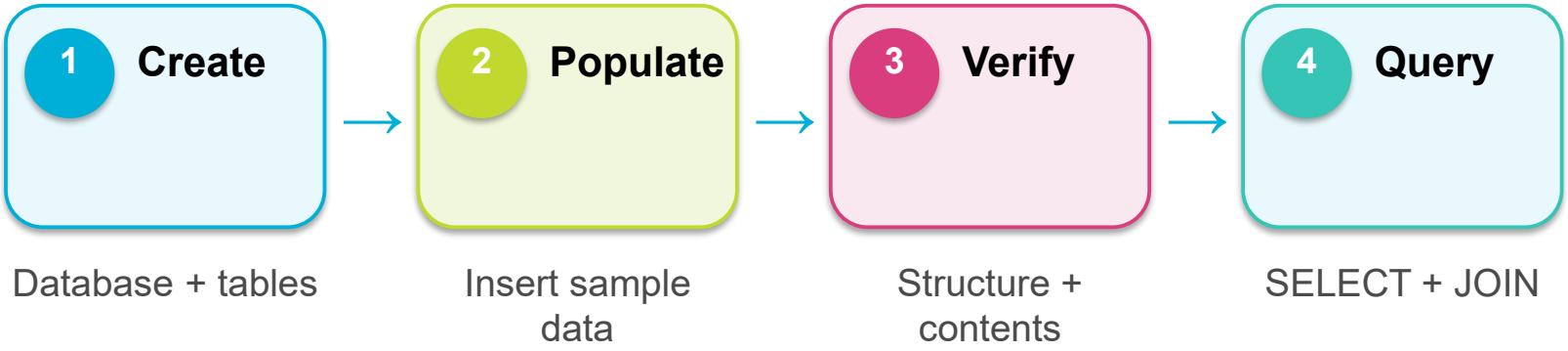
AdCampaignID	ModusID	BudgetPctg
111	1	50%
111	2	50%
222	1	60%
222	3	30%
222	5	10%
333	3	80%
333	4	20%
555	3	100%

Insertion Anomaly Resolved:
New campaign modus 7 inserted without inserting an actual campaign using the new modus 7.

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Practical Workflow

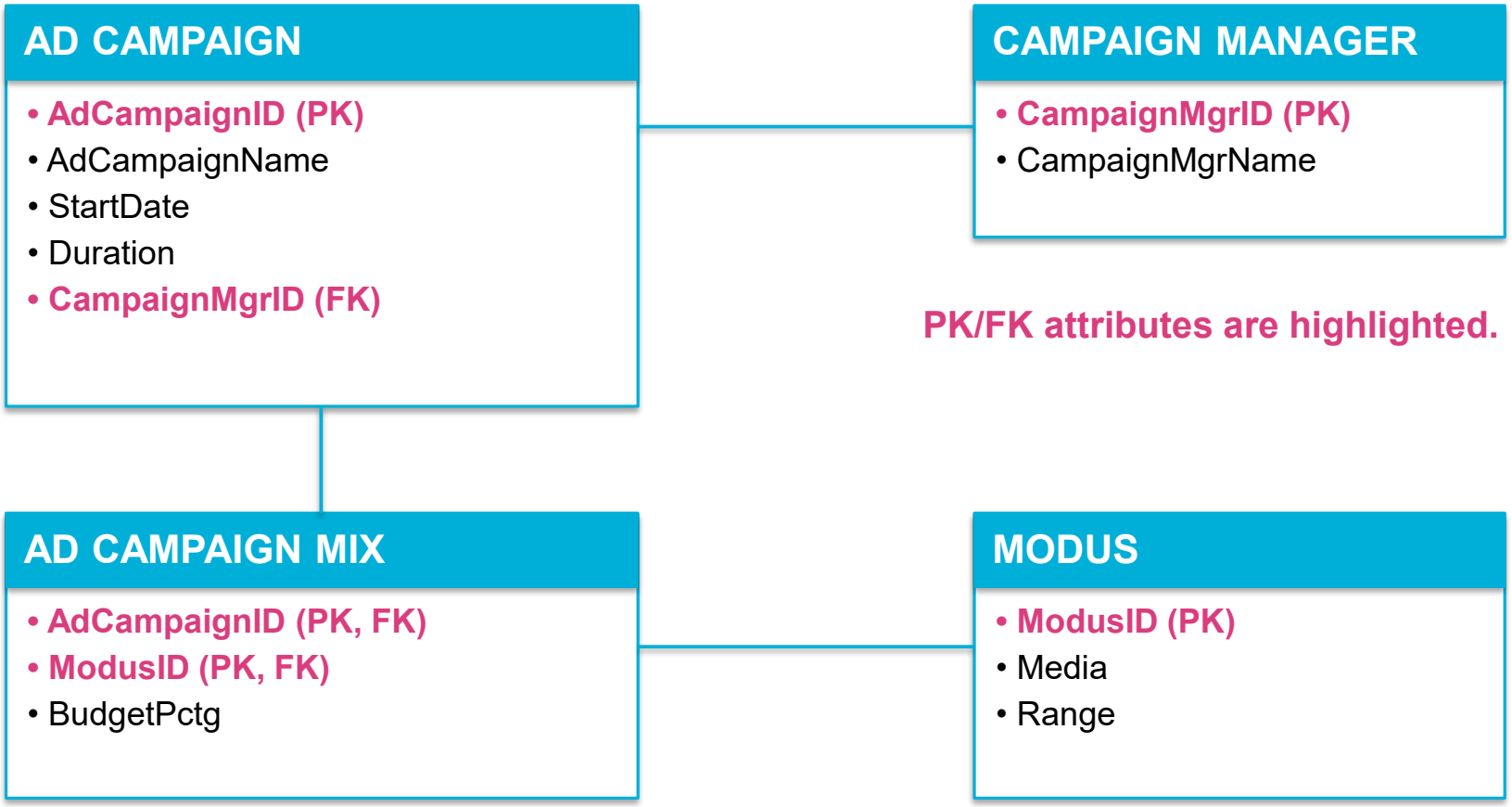
Normalization theory is assumed from the lecture sessions.



Goal: Implement the 3NF design correctly and prove that the relations work together.

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3NF Database Structure



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AD CAMPAIGN

AdCampaignID	AdCampaignName	StartDate	Duration	CampaignMgrID
111	SummerFun20	6.6.2020	12 days	CM100
222	SummerZing20	6.8.2020	30 days	CM101
333	FallBall20	6.9.2020	12 days	CM102
444	AutmnStyle20	6.9.2020	5 days	CM103
555	AutmnColors20	6.9.2020	3 days	CM100

Primary Key: AdCampaignID **Foreign Key:** CampaignMgrID

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CAMPAIGN MANAGER and MODUS

CAMPAIGN MANAGER

CampaignMgrID	CampaignMgrName
CM100	Roberta
CM101	Sue
CM102	John
CM103	Nancy

Primary Key: CampaignMgrID

MODUS

ModusID	Media	Range
1	TV	Local
2	TV	National
3	Radio	Local
4	Radio	National
5	Print	Local
6	Print	National
7	Internet	National

Primary Key: ModusID

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AD CAMPAIGN MIX

AdCampaignID	ModusID	BudgetPctg
111	1	50%
111	2	50%
222	1	60%
222	3	30%
222	5	10%
333	3	80%
333	4	20%
444	6	100%
555	3	100%

Composite PK: (AdCampaignID, ModusID)

Foreign Keys: AdCampaignID, ModusID

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MySQL Implementation

Follow the steps in order to avoid foreign-key errors.

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Step 1: Create and Select the Database

```
CREATE DATABASE advertise;

USE advertise;

SHOW DATABASES;
```

Parent tables first → child tables afterwards.

Create tables in this order

1. **CampaignManager**
2. **Modus**
3. **AdCampaign**
4. **AdCampaignMix**

Steps 2–3: Create the Parent Tables

CampaignManager

```
CREATE TABLE CampaignManager (
  CampaignMgrID VARCHAR(5) PRIMARY KEY,
  CampaignMgrName VARCHAR(50) NOT NULL
);
```

Modus

```
CREATE TABLE Modus (
  ModusID INT PRIMARY KEY,
  Media VARCHAR(20) NOT NULL,
  `Range` VARCHAR(20) NOT NULL
);
```

CampaignMgrID and ModusID uniquely identify rows in their tables.

`Range` is enclosed in backticks because RANGE can be interpreted as a MySQL keyword.

Steps 4–5: Create the Child Tables

AdCampaign

```
CREATE TABLE AdCampaign (
  AdCampaignID INT PRIMARY KEY,
  AdCampaignName VARCHAR(50) NOT NULL UNIQUE,
  StartDate DATE NOT NULL,
  Duration INT NOT NULL,
  CampaignMgrID VARCHAR(5) NOT NULL,
  FOREIGN KEY (CampaignMgrID)
    REFERENCES CampaignManager(CampaignMgrID)
);
```

AdCampaignMix

```
CREATE TABLE AdCampaignMix (
  AdCampaignID INT,
  ModusID INT,
  BudgetPctg DECIMAL(5,2) NOT NULL,
  PRIMARY KEY (AdCampaignID, ModusID),
  FOREIGN KEY (AdCampaignID)
    REFERENCES AdCampaign(AdCampaignID),
  FOREIGN KEY (ModusID)
    REFERENCES Modus(ModusID)
);
```

Duration is stored as a number (e.g., 12 rather than “12 days”).

Steps 6–7: Verify the Database Structure

```
SHOW TABLES;
```

Expected tables

AdCampaign
AdCampaignMix
CampaignManager
Modus

```
DESCRIBE CampaignManager;
DESCRIBE Modus;
DESCRIBE AdCampaign;
DESCRIBE AdCampaignMix;
```

Observe

- column names
- data types
- key columns
- NULL / NOT NULL settings

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Steps 8–9: Insert Parent-Table Data

CampaignManager

```
INSERT INTO CampaignManager
(CampaignMgrID, CampaignMgrName)
VALUES
('CM100', 'Roberta'),
('CM101', 'Sue'),
('CM102', 'John'),
('CM103', 'Nancy');
```

Modus

```
INSERT INTO Modus
(ModusID, Media, `Range`)
VALUES
(1, 'TV', 'Local'),
(2, 'TV', 'National'),
(3, 'Radio', 'Local'),
(4, 'Radio', 'National'),
(5, 'Print', 'Local'),
(6, 'Print', 'National'),
(7, 'Internet', 'National');
```

Insert parent records before rows that reference them.

Steps 10–11: Insert Campaign Data

AdCampaign

```
INSERT INTO AdCampaign
(AdCampaignID, AdCampaignName, StartDate,
   Duration, CampaignMgrID)
VALUES
(111, 'SummerFun20', '2020-06-06', 12, 'CM100'),
(222, 'SummerZing20', '2020-08-06', 30, 'CM101'),
(333, 'FallBall20', '2020-09-06', 12, 'CM102'),
(444, 'AutmnStyle20', '2020-09-06', 5, 'CM103'),
(555, 'AutmnColors20', '2020-09-06', 3, 'CM100');
```

AdCampaignMix

```
INSERT INTO AdCampaignMix
(AdCampaignID, ModusID, BudgetPctg)
VALUES
(111, 1, 50), (111, 2, 50),
(222, 1, 60), (222, 3, 30),
(222, 5, 10),(333, 3, 80),
(333, 4, 20),(444, 6, 100),
(555, 3, 100);
```

Check each insert with SELECT * FROM <table>;

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Step 12: View the Contents of All Tables

```
SELECT * FROM CampaignManager;
SELECT * FROM Modus;
SELECT * FROM AdCampaign;
SELECT * FROM AdCampaignMix;
```

Compare with the supplied 3NF relations

1. **CampaignManager**
2. **Modus**
3. **AdCampaign**
4. **AdCampaignMix**

Matching rows and keys confirm that the database was populated correctly.

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Steps 13–14: Join Related Tables

Campaigns + Managers

```
SELECT
  AdCampaign.AdCampaignID,
  AdCampaign.AdCampaignName,
  CampaignManager.CampaignMgrName
FROM AdCampaign
JOIN CampaignManager
  ON AdCampaign.CampaignMgrID =
     CampaignManager.CampaignMgrID;
```

Campaigns + Advertising Modes

```
SELECT
  AdCampaign.AdCampaignName,
  Modus.Media,
  Modus.`Range`,
  AdCampaignMix.BudgetPctg
FROM AdCampaign
JOIN AdCampaignMix
  ON AdCampaign.AdCampaignID =
     AdCampaignMix.AdCampaignID
JOIN Modus
  ON AdCampaignMix.ModusID = Modus.ModusID;
```

A campaign may appear multiple times because it may use multiple advertising modes.

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Step 15: Reconstruct the Original Relation

```
SELECT
  ac.AdCampaignID,
  ac.AdCampaignName,
  ac.StartDate,
  ac.Duration,
  ac.CampaignMgrID,
  cm.CampaignMgrName,
  m.ModusID,
  m.Media,
  m.`Range`,
  acm.BudgetPctg
FROM AdCampaign ac
JOIN CampaignManager cm
  ON ac.CampaignMgrID = cm.CampaignMgrID
JOIN AdCampaignMix acm
  ON ac.AdCampaignID = acm.AdCampaignID
JOIN Modus m
  ON acm.ModusID = m.ModusID
ORDER BY ac.AdCampaignID, m.ModusID;
```

Why this matters

The information is stored in separate normalized tables, but JOIN operations can reconstruct the complete campaign data whenever it is needed.

This verifies that the decomposition has not lost the original information.

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Steps 16–17: Test Database Constraints

Primary-key test

```
INSERT INTO CampaignManager
VALUES ('CM100', 'Another Manager');
```

Expected: rejected

Duplicate entry 'CM100' for key 'PRIMARY'

Primary key → prevents duplicate identifiers.

Foreign-key test

```
INSERT INTO AdCampaign
VALUES (777, 'TestCampaign',
        '2020-10-01', 10, 'CM999');
```

Expected: rejected

CM999 does not exist in CampaignManager.

Foreign key → prevents invalid references.

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